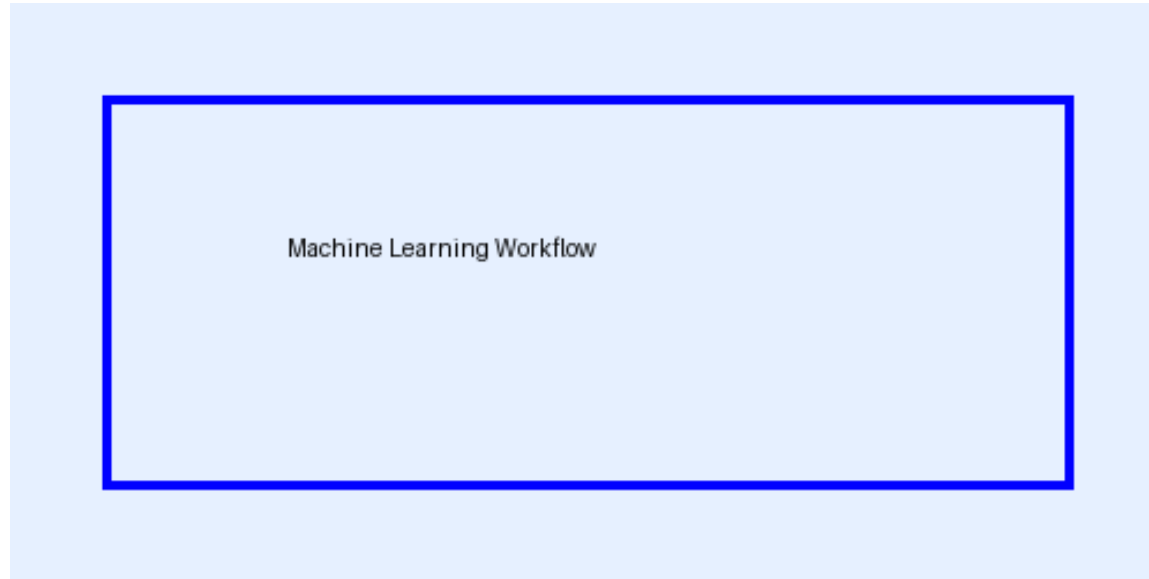


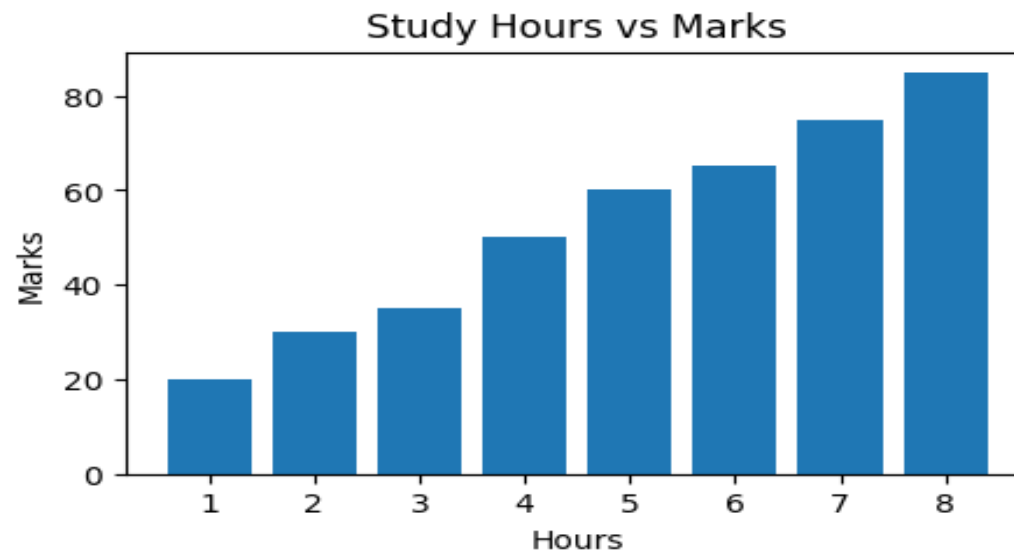
TASK 2 - MACHINE LEARNING

Predict Student Marks using Linear Regression



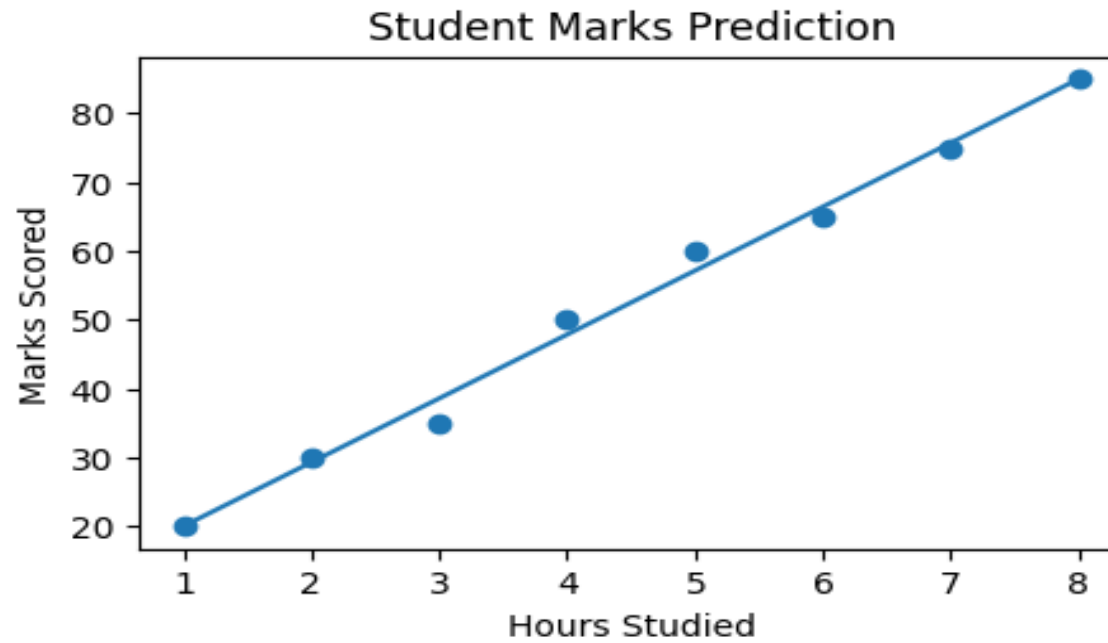
Project Objective

- Predict student marks based on study hours.
- Use Python, Pandas, Matplotlib, and Scikit-learn.
- Apply Linear Regression for prediction.
- Evaluate model performance with accuracy metrics.



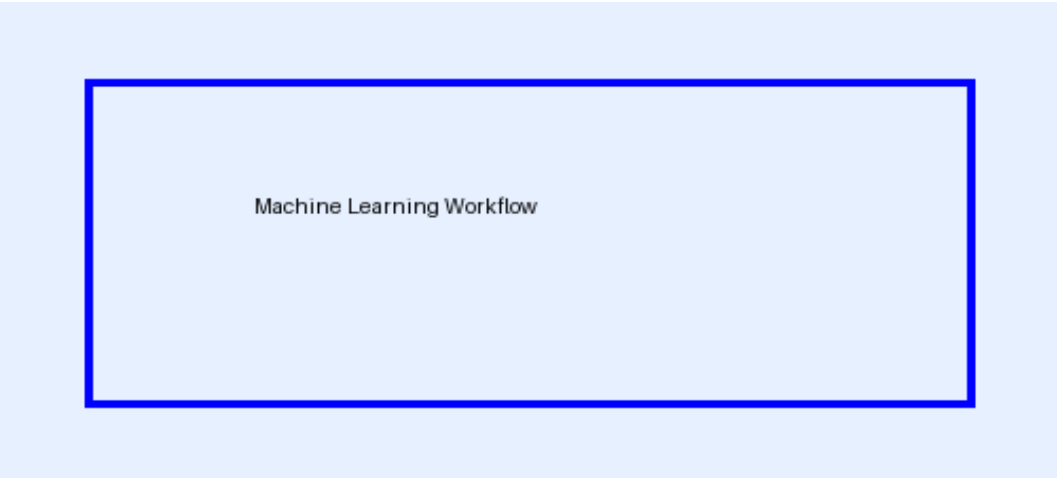
Data Visualization

The graph below shows the relationship between hours studied and marks scored. A positive trend indicates that marks improve as study hours increase.



Model Training Process

1. Load dataset using Pandas.
2. Split data into training and testing sets.
3. Train Linear Regression model.
4. Predict marks for new study hours.
5. Evaluate accuracy using MAE and R^2 Score.



Machine Learning Workflow

The diagram consists of a large light blue rectangle containing a smaller rectangle with a thick blue border. The text "Machine Learning Workflow" is centered within the inner rectangle.

Conclusion

- Linear Regression is simple and effective for prediction tasks.
- The model helps estimate marks from study hours.
- This project demonstrates a real-world machine learning workflow.
- Useful for beginners to understand supervised learning concepts.

